

Deep Generative Models

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Problem Set 1

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1. Properties of Gaussian Distributions

Linear Transformation on Gaussians

If $\vec{X} \in \mathbb{R}^d$ is a gaussian, $X \sim \mathcal{N}(\vec{\mu}, \Sigma)$, then for a fixed matrix $A_{m \times d}$ and $\vec{b} \in \mathbb{R}^d$ we have :